姓名________ 学号________ 学院________ 专业班级________ 座位号________

（密封线内不答题） 密 封 线

诚信应考，考试作弊将带来严重后果！

华南理工大学本科生期末考试

2024-2025-2 学期章节试卷

《Calculus Chapter Review-Partial Derivative》

注意事项：1. 开考前请将密封线内各项信息填写清楚；
2. 所有答案请直接答在试卷上；
3. 考试形式：闭卷；
4. 本试卷共 5 大题，满分 50 分，考试时间 50 分钟。

1, Find $\frac{\partial z}{\partial x}$, if $x^3 e^{y+z} - y\sin(x-z) = 0$

2, Find maximum-minimum of $f(x,y,z) = -x+2y+2z$ on ellipse, $x^2+y^2=2,\ y+2z=1$.

3, Show the limit does not exist. $\lim\limits_{(x,y,y)\to(0,0,0)} \frac{xyz}{x^3+y^3+z^3}$

4, Find curvature curve that is intersection of $x^2+y^2+z^2=1$ and $3x+2y+6z=0$

5，Show $\frac{\partial^2 f}{\partial x\partial y}(0,0) \neq \frac{\partial^2 f}{\partial y\partial x}(0,0)$, if $f(x,y)=\begin{cases} xy\frac{x^2-y^2}{x^2+y^2}, & (x,y)\neq(0,0) \\ 0, & (x,y)=(0,0) \end{cases}$

7, Find the curvature for the cardioid $r = 1+\cos\theta,\ at\ \theta=0$

8，$z=z(x,y)$ *is determined by* $F(x+y,y+z,z+x)=0$ *Find* $\frac{\partial^2 z}{\partial x\partial y}$

9，*Find tangent line equation and normal plane of curve* $\begin{cases} z=x^2+y^2 \\ 2x^2+2y^2=z^2 \end{cases}$ *at* $(1,1,2)$;

10，$z = xf(\frac{y}{x}) + 2yf(\frac{x}{y})$ *Find* $x\frac{\partial^2 z}{\partial x^2} + y\frac{\partial^2 z}{\partial x\partial y}$

If $f(x,y)=\begin{cases} (x^2+y^2)\sin\frac{1}{x^2+y^2}, & x^2+y^2\neq 0 \\ 0, & x^2+y^2=0 \end{cases}$

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(1) $f(x,y)$ is *continue or not at* $(0,0)$

(2) $\frac{\partial f(x,y)}{\partial x}, \frac{\partial f(x,y)}{\partial y}$ *are continue or not at* $(0,0)$